



# **DESIGN THINKING**

## **& FABRICATION LABORATORY**

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**Design Thinking** is a human-centred, iterative approach to innovation that puts *people first* and turns real problems into workable solutions.

- ▶ Blends **empathy**, **creativity** and **rationality**.
- ▶ Moves from *understanding people* to *building and testing* tangible solutions.
- ▶ The **Fabrication Laboratory** turns ideas into physical prototypes using digital fabrication tools.

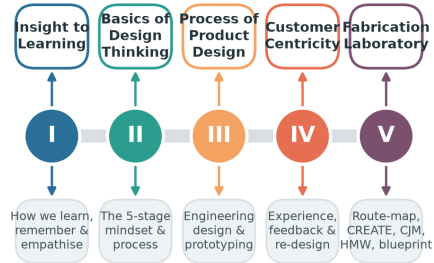
**Course structure:** five units of 15 hours each, combining theory with hands-on making.

## Learning outcomes

By the end of the course you will be able to:

- ▶ Understand how people learn and empathise.
- ▶ Apply the 5-stage design-thinking process.
- ▶ Design, prototype and test products.
- ▶ Map customer journeys and service blueprints.
- ▶ Build solutions in a fabrication lab.

## Course Roadmap — Five Units, One Journey



*15 hours per unit · 75 hours total · theory + hands-on fabrication*

*Five units, one continuous journey from insight to fabrication.*

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- 1 An Insight to Learning
  - 2 Basics of Design Thinking
  - 3 Process of Product Design
  - 4 Design Thinking & Customer Centricity
  - 5 Fabrication Laboratory



# An Insight to Learning

Learning, memory, emotion & empathy

15 hours

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Learning is the process of acquiring **knowledge**, **skills** and **attitudes** through experience, study or teaching.

- ▶ **Cognitive** — thinking, understanding, problem solving.
- ▶ **Affective** — emotions, values, motivation.
- ▶ **Psychomotor** — physical / hands-on skills.

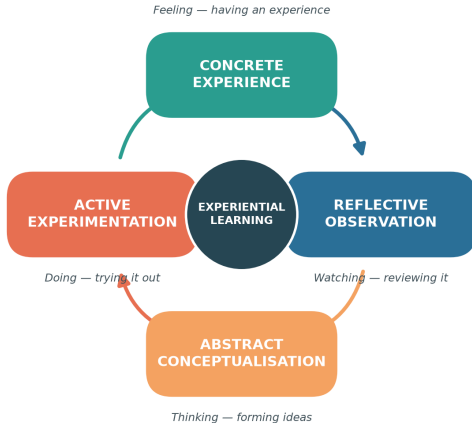
Effective designers are *lifelong learners*: they observe, reflect, conceptualise and experiment — exactly the loop Kolb described.

## Why it matters for design

Understanding *how humans learn* helps us design products, instructions and experiences that people can pick up quickly and use intuitively.

## Kolb's Experiential Learning Cycle

*Learning as a continuous four-stage loop*



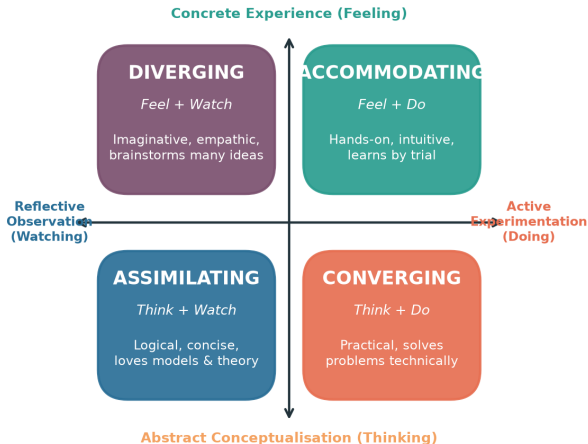
Learning is a **continuous cycle** of four stages:

- 1 **Concrete Experience** — do / feel.
- 2 **Reflective Observation** — review.
- 3 **Abstract Conceptualisation** — conclude.
- 4 **Active Experimentation** — plan & try.

Effective learning happens when we travel *all the way round* the loop — repeatedly.

## Kolb's Four Learning Styles

*Formed by how learners grasp and transform experience*



Combining *how we grasp* and *how we transform* experience gives four styles:

- ▶ **Diverging** — imaginative, many ideas.
- ▶ **Accommodating** — hands-on, intuitive.
- ▶ **Assimilating** — logical, loves theory.
- ▶ **Converging** — practical problem-solver.

Teams work best when *all four styles* are present.



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## How we assess

- ▶ **Learning Style Inventory (LSI)** questionnaires.
- ▶ Self-reflection journals.
- ▶ Observation of behaviour in tasks.

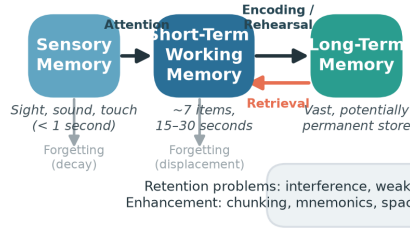
## How we interpret

- ▶ No style is *better* — each has strengths.
- ▶ Match **teaching methods** to styles.
- ▶ Encourage learners to stretch into weaker modes.

## Application with peers

In group projects, deliberately mix styles so ideation, analysis, building and testing are all covered.

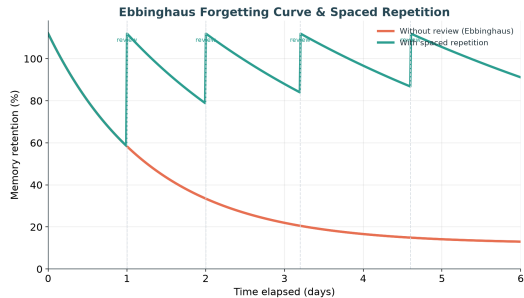
## The Memory Process — Information-Processing Model



*The information-processing model: sensory → short-term → long-term memory.*

## Why do we forget?

- ▶ **Decay** — traces fade without use.
- ▶ **Interference** — old & new memories clash.
- ▶ **Retrieval failure** — no cue to recall.
- ▶ **Weak encoding** — shallow processing.



The **Ebbinghaus curve** shows rapid forgetting — but each review flattens the curve.

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- ▶ **Chunking** — group items into meaningful units.
  - ▶ **Mnemonics** — acronyms, rhymes, imagery.
  - ▶ **Spaced repetition** — review at intervals.
  - ▶ **Elaboration** — connect to prior knowledge.
  - ▶ **Dual coding** — words + visuals together.
  - ▶ **Active recall** — test yourself, don't re-read.
  - ▶ **Method of loci** — memory "palace".
  - ▶ **Sleep & rest** — consolidate memories.

## Design tip

Good product design *reduces the memory load* on the user — clear labels, consistent layouts and helpful cues.

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Emotions shape how people **experience** and **remember** products.

- ▶ **Emotion** — an internal feeling (joy, frustration, trust).
- ▶ **Experience** — the felt journey of using something.
- ▶ **Expression** — how emotion is shown & communicated.

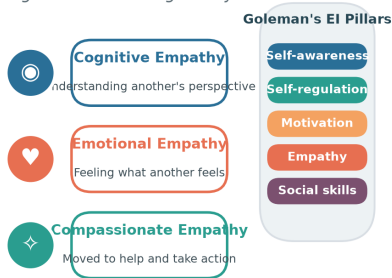
Peak emotions (delight or annoyance) leave the strongest memories — the *peak-end rule*.

## Emotional design (Norman)

- ▶ **Visceral** — first look & feel.
- ▶ **Behavioural** — pleasure of use.
- ▶ **Reflective** — meaning & self-image.

## Emotions, Empathy & Emotional Intelligence

*Understanding emotions is the gateway to human-centred design*



*Assessing empathy with peers → empathy maps, shadowing, and active-listening interviews*

**Empathy** — the ability to understand and share another's feelings — is the *heart of design thinking*. **Assessing empathy with peers**

- ▶ Empathy maps (says / thinks / does / feels).
- ▶ Shadowing & observation.
- ▶ Active-listening interviews.



# Basics of Design Thinking

The mindset, the process, the tools

15 hours

## Definition

A **human-centred**, iterative problem-solving approach that seeks to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test.

## Need for design thinking

- ▶ Problems are **complex & ill-defined**.
- ▶ Users' real needs are often *hidden*.
- ▶ Reduces the risk of building the wrong thing.

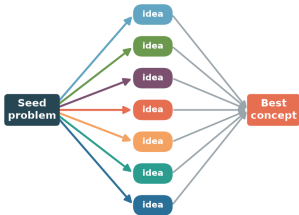
## Objectives

- ▶ Keep the **user at the centre**.
- ▶ Encourage **creativity & collaboration**.
- ▶ **Fail early**, learn cheaply.
- ▶ Turn insight into tangible value.



## Brainstorming — Divergent then Convergent Thinking

*Rules: quantity over quality · build on others · one conversation · stay visual*



**DIVERGE** — generate many ideas, defer judgement, go wild

**CONVERGE** — group, evaluate & select

Creative work alternates two modes:

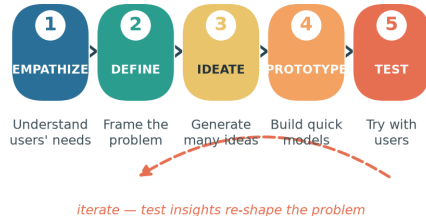
- ▶ **Diverge** — generate many ideas, defer judgement.
- ▶ **Converge** — group, evaluate & select.

### Brainstorming rules

- ▶ Go for quantity.
- ▶ Build on others' ideas.
- ▶ One conversation at a time.
- ▶ Stay visual; welcome wild ideas.

## The Five Stages of Design Thinking

*Non-linear and iterative — teams loop back as they learn*



*Empathize → Define → Ideate → Prototype → Test — iterative, not linear.*

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1. **Empathize** Observe and engage with users; conduct interviews and immerse yourself in their world. *Example:* shadowing commuters to learn how they use a ticket machine.
  2. **Define** Synthesise observations into a clear, user-centred **problem statement** (a Point-of-View). *Example:* "Busy commuters need a faster way to buy tickets because queues cause stress."
  3. **Ideate** Generate a wide range of ideas through brainstorming, sketching and mind-mapping — *quantity over quality* first.

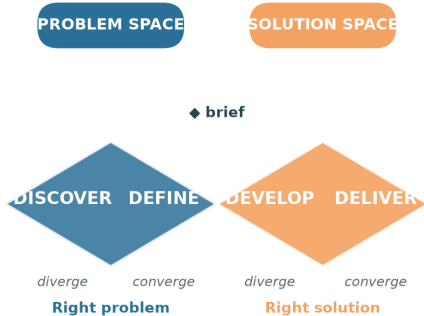
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- 4. **Prototype** Build quick, inexpensive, scaled-down versions to explore ideas. *Example:* a paper mock-up of the ticket-machine screen.
  - 5. **Test** Try prototypes with real users, gather feedback and refine. Testing often *re-defines the problem* — sending you back to earlier stages.

### Being ingenious

The stages are a *flexible toolkit*, not a rigid recipe. Teams loop, skip and revisit stages as they learn.

## The Double Diamond

*Alternating divergent (explore) and convergent (focus) thinking*

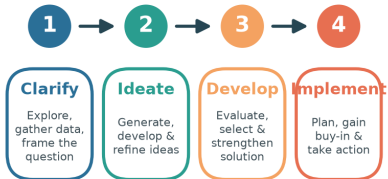


A popular map of the design process:

- ▶ **Discover & Define** — find the *right problem*.
- ▶ **Develop & Deliver** — build the *right solution*.

Each diamond widens (diverge) then narrows (converge).

## Creative Problem-Solving Process



Testing creative problem solving: judge originality, fluency, flexibility & elaboration of ideas

Creative problem solving moves through four stages: **Clarify**, **Ideate**, **Develop** and **Implement**. Testing creative problem

solving — judge ideas on:

- ▶ *Fluency* — number of ideas.
- ▶ *Flexibility* — variety of ideas.
- ▶ *Originality* — novelty.
- ▶ *Elaboration* — detail & refinement.

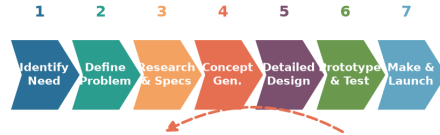


# Process of Product Design

Engineering design, prototyping & testing

15 hours

## Engineering Product-Design Process



iterate on test results

*design-thinking approach keeps the user in the loop at every stage*

*A structured flow from identifying a need to manufacture & launch.*



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## Traditional engineering

- ▶ Requirements defined up front.
- ▶ Linear, specification-driven.
- ▶ User feedback comes late.

## Design-thinking infused

- ▶ **User needs** continuously explored.
- ▶ **Iterative** — prototype & test early.
- ▶ Feedback loops at every stage.

## Best in class

Great products (e.g. intuitive smartphones, ergonomic tools, well-designed appliances) combine *function*, *form* and *delight* — form follows function *and* feeling.

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A **prototype** is an early, testable model of a product used to *explore ideas* and *learn* before committing to production. **Why prototype?**

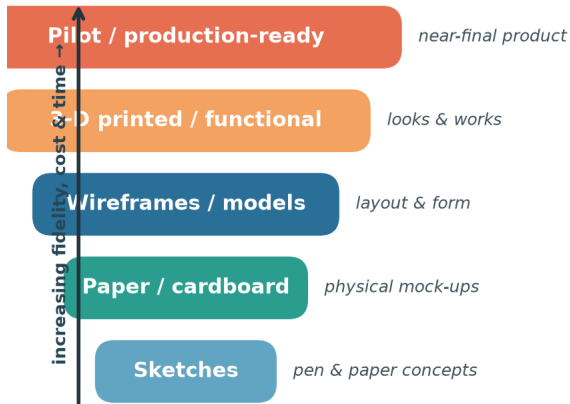
- ▶ Make ideas **tangible**.
- ▶ Test assumptions cheaply.
- ▶ Communicate with stakeholders.
- ▶ Fail fast, improve faster.

## Types of prototype

- ▶ *Low-fidelity* — sketches, paper, cardboard.
- ▶ *Mid-fidelity* — wireframes, models.
- ▶ *High-fidelity* — 3-D printed, functional.

## Prototype Fidelity Ladder

*From cheap & rough to refined & functional*



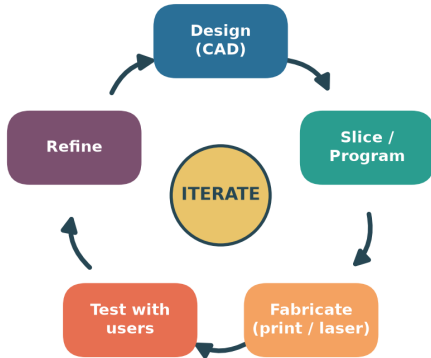
Start **low** and cheap, climb only as questions demand.

- ▶ Low fidelity answers *"is this the right idea?"*
- ▶ High fidelity answers *"does it work & feel right?"*

Each rung costs more time and money — so *learn cheaply first*.

## Rapid Prototyping Cycle

*Fail fast, learn faster — CAD to physical part in hours*



*Sample: laser-cut acrylic housing → test-group marketing → refine*

**Rapid prototyping** uses digital fabrication (3-D printing, laser cutting, CNC) to go from **CAD** to a physical part in *hours*. **Testing & sample marketing**

- ▶ Test the prototype with a small **test group**.
- ▶ Gather reactions before scaling up.
- ▶ Refine, then repeat the loop.



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# Design Thinking & Customer Centricity

Experience, feedback & re-creation

15 hours

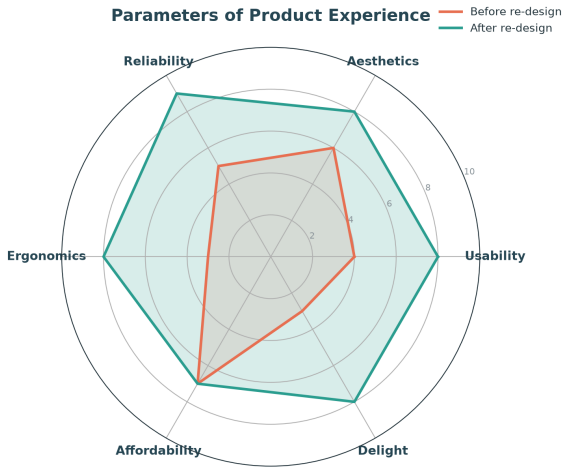
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Real customers face everyday frustrations that design thinking can solve:

- ▶ Products that are **hard to assemble** or use.
- ▶ Confusing instructions & interfaces.
- ▶ Poor **ergonomics** causing discomfort.
- ▶ Long waits and clunky service.
- ▶ Features nobody asked for.
- ▶ A gap between the *promise* and the *experience*.

## Design thinking to the rescue

By empathising with customers we uncover these pains and re-design the product experience around them.

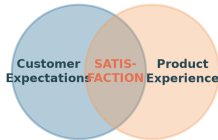


Product experience is multi-dimensional:

- Usability & Reliability
- Aesthetics & Ergonomics
- Affordability & Delight

Thoughtful re-design lifts every axis — turning a merely functional product into one customers *love*.

## Aligning Customer Expectations with the Product



*Gap too wide → disappointment · aligned → loyalty & advocacy*

**Satisfaction** lives where *expectations* meet *experience*.

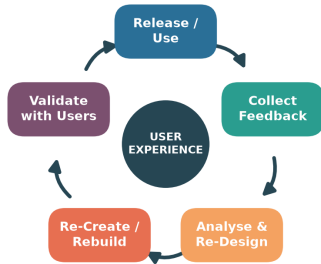
- ▶ Gap too wide → disappointment.
- ▶ Well aligned → loyalty & advocacy.

Manage expectations honestly *and* raise the experience to close the gap.



## Feedback → Re-Design → Re-Create Loop

*Continuous improvement centred on user experience*



*Address ergonomic challenges · close the gap between expectation and experience*

The **feedback loop** keeps a product improving:

- 1 Release & use.
- 2 Collect feedback.
- 3 Analyse & re-design.
- 4 Re-create / rebuild.
- 5 Validate with users.

Focus relentlessly on **user experience** and address ergonomic challenges each cycle.

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## User-focused design

- ▶ Design *with* users, not just for them.
- ▶ Fit the product to the human — **ergonomics**.
- ▶ Test with real tasks in real contexts.

## Toward the final product

- ▶ Converge feedback into refinements.
- ▶ Validate safety, comfort & usability.
- ▶ Ship — then keep listening.

## Remember

A "final" product is really the *best current version* — the loop never truly ends.



# Fabrication Laboratory

The route map: survey to service blueprint

15 hours

## Fab-Lab Route Map — From Survey to Service Blueprint



A step-by-step recipe for the lab project:

- 1 Conduct surveys.
- 2 Identify a problem.
- 3 Frame a problem statement.
- 4 Apply the **CREATE** tool.
- 5 Draw the product / system.
- 6 Build a Customer Journey Map.
- 7 Frame 2–3 **HMW** questions.
- 8 Design a Service Blueprint.

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## Conduct surveys

- ▶ Individually or as a group.
- ▶ Observe, interview, question.
- ▶ Look for **real, felt pains**.

**Identify a problem** worth solving.

## Framing a problem statement

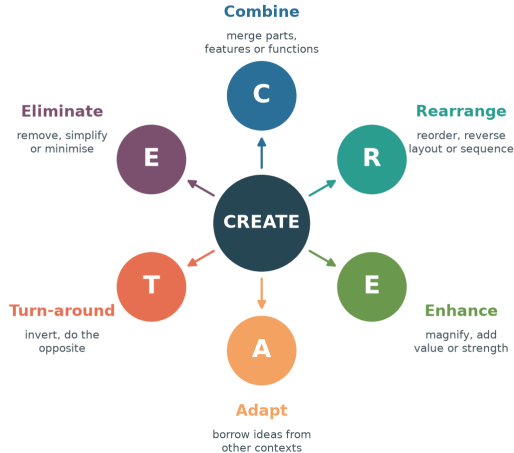
A good statement is **user-centred**, **clear** and **actionable**:

*"[User] needs [need] because [insight]."*

Avoid baking in a solution — describe the *problem*.

## The CREATE Ideation Tool

*Six trigger words to transform an existing product*



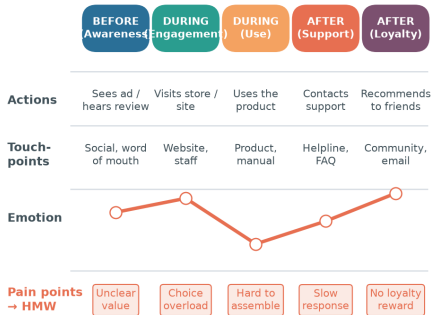
**CREATE** is a set of *trigger words* to transform an existing product:

- ▶ **Combine**    **Rearrange**
- ▶ **Enhance**    **Adapt**
- ▶ **Turn-around**    **Eliminate**

Apply each trigger to your problem, then **draw the product / system** that results.

## Customer Journey Map (CJM)

*Mapping the experience — before, during & after*

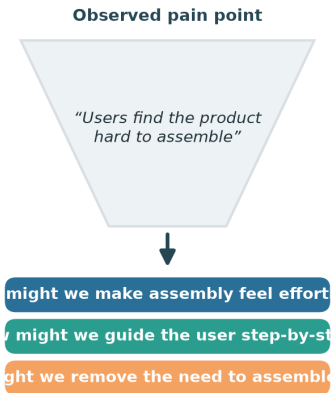


*Touch-points feed the Service Blueprint · pain points become How-Might-We questions*

*Map actions, touch-points, emotions & pain points — before, during & after.*

## How Might We (HMW) Questions

*Turning pain points into springboards for ideation*



Turn each **pain point** from the CJM into an optimistic **"How Might We..."** question.

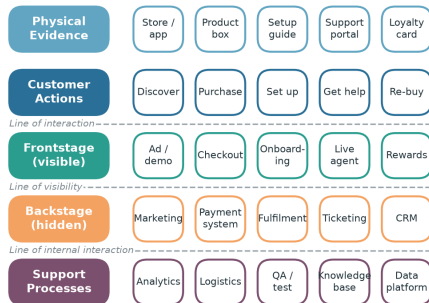
- ▶ Broad enough to *inspire* ideas.
- ▶ Narrow enough to *act on*.
- ▶ Frame 2–3 for your mock scenario.

*Frame 2–3 HMW questions — broad enough to inspire, narrow enough to act*



## Service Blueprint

*Layering the customer journey with what happens behind the scenes*



*Layer the journey with front-stage, back-stage & support processes; identify touch-points.*

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- ▶ **Unit I** — learn, remember, empathise.
  - ▶ **Unit II** — the design-thinking process.
  - ▶ **Unit III** — design & prototype products.
  - ▶ **Unit IV** — centre on the customer.
  - ▶ **Unit V** — build & blueprint in the lab.

## The golden thread

Every unit returns to one idea: **empathise with people, then iterate** — observe, frame, ideate, prototype, test, and repeat until the experience truly serves the user.

# Thank You

Design Thinking & Fabrication Laboratory

Empathise → Define → Ideate → Prototype → Test